Mathematical and computational supplement to A certificate-based audit of Wuebben's proposed exotic $S^2 \times S^2$ construction

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August 2026

Abstract

This supplement preserves the implementation inventory, complete downstream dependency exposition, trust boundary, data pins, calibration cases, and historical exports supporting the main paper. None of the material here changes the logical status stated there: finite certificates are replayable, the audit-defined manifold V_* is simply connected, the comparison $V_* \cong V$ is conditional on the four source-identification assumptions S1–S4, and the downstream manifold conclusions are additionally relative to named external results.

How to use this supplement

The main paper contains the mathematical certificate specification and the proof of the audit-model identification, followed by the explicitly conditional comparison with Wuebben. Section 1 below records the concrete certificate batches and replay architecture. Appendix A gives the full proposition-by-proposition downstream chain; Appendix B records provenance, trust, and data availability; Appendix C preserves calibrations and historical computations. The machine-readable ledger and every bound digest remain normative for exact inventories.

1 Certificate inventory and replay

1.1 The four convention-specialized groups

For each of the four $n = 0$ entries defined in the finite theorem of the main paper, KBMAG produced a complete rewriting system in which all three numbered generators reduce to the identity. Completion is only the discovery method: each verdict is exported as a derivation DAG and proved by the Certificate Soundness Theorem in the main paper. The certificate sizes are shown below.

system	signs	retained records
n=0	$(-,-)$	3620
n=0	$(-,+)$	3803
n=0	$(+,-)$	2470
n=0	$(+,+)$	6672

Retained derivation-DAG records over the sealed presentation. Each certificate verifies triviality from a fresh clone.

An earlier four-generator export, four adjacent $n = 1$ controls, calibration examples, deliberate corruptions, and the full framing-scan history are summarized in Section C. None carries logical load for the fixed $n = 0$ target.

1.2 Verifier architecture

Knuth–Bendix completion (KBMAG 1.5.9 [25], run under GAP 4.11.1 [26]) is used only as an *untrusted certificate generator*. A patched build of KBMAG records the ancestry of every equation it derives; the patch is distributed as `verification/luttinger/kbmag-proof/kbfns-proof.patch`. An untrusted compiler prunes that history to the dependency cone of the final identity rules and emits a certificate of the form in the Certificate Soundness Theorem in the main paper, binding the input presentation by SHA-256. The certified set is enforced, not assumed: both checkers reject a batch that verifies any case twice, and in full-inventory mode — the documented invocation — require the batch to be exactly the input's eight fillings, one file named by each case slug, over a source of the stated shape (three generators and 78 complement relators for the sealed chain, four and 95 for the earlier export) with two filling relators per case; the manifest generator refuses to hash a certificate directory that is not exactly one file per filling. A successful replay therefore verifies exactly the four n=0 sign pairs and the four n=1 controls. A small standalone checker then re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, tidying change, and final generator-to-identity rule; it neither imports nor runs KBMAG. A second checker, separately implemented in Ruby from the same specification, re-verifies all retained derivation records against the same hash-bound certificates and accepts, importing neither the first checker nor any project group code. All certificates, the exported rewriting systems, and a hash manifest replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KBMAG independently of the certificate path.

The transport from the raw simplicial presentation to the presentation whose fillings are certified is sealed in the same sense. The canonical seeded run serializes the raw complement presentation (99,863 generators, 321,702 relators) together with its 89 tracked peripheral and coordinate words, records the 99,860 elementary eliminations that reduce it to the 3-generator, 78-relator presentation, and freezes that output. Two standalone checkers — one in Python, one in Ruby, importing neither the geometry nor the simplifier — replay the eliminations from those three frozen files alone: a step is accepted only if the eliminated generator occurs exactly once in the named current relator and the recorded replacement is the one that relator implies, input and output are bound by SHA-256, and the renumbered result must equal the frozen presentation relator by relator and tracked word by tracked word. Both reject corrupted-input, omitted-step, altered-substitution, and altered-output controls. The chain from the frozen raw complex to the eight triviality verdicts (39,163 retained derivation records over the sealed presentation) therefore replays end to end from committed files. The design principle throughout is that certificates are checked, not trusted: search and completion tools are untrusted generators, and every computational claim rests on a replayable artifact validated by small independent code.

The 78-relator input has $H_1 = \mathbb{Z}^2$, and each filling pair kills its abelianization; triviality rather than mere perfectness is carried by the derivation certificates. The negative controls and calibration examples in Section C show that the pipeline also produces nontrivial groups and that both checkers reject altered records and incomplete inventories. Those controls test implementation behavior; they are not substitutes for the soundness argument above.

A From simple connectivity to the three theorems

The deductions from $\pi_1(V) = 1$ to Theorems A, B, C occupy Part I of [1], whose Theorem 1.2 reduces them to the simple connectivity of V. Throughout this appendix, S1–S4 of the main paper are standing assumptions, so its audit-defined V_* has been identified with this V. This section carries the subsequent deductions through as an explicit dependency chain. The chain is itself a machine artifact: a script records every item as one of four kinds — an *external theorem* stated with its hypotheses and source; a *certificate* of this repository, bound by SHA-256; a *computed* fact, executed by the script; or a *step*, a deduction from earlier items — and freezes the result as a certificate. A second script, separately implemented in Ruby, re-derives every computed fact from scratch, checks that every citation resolves and that the dependency graph is acyclic, verifies that each of the three conclusions depends on the finite theorem of the main paper, and checks every evidence file against its recorded digest. The chain has 25 external theorems, 3 certificates, 15 computed facts, and 16 steps; its prose form is in the repository, and what follows is the mathematics, with the inputs first. Both scripts in this downstream pair were produced by Claude Fable 5; independence here likewise means separate implementations, not cross-model authorship.

External inputs

Elementary algebraic topology enters as: the Seifert–van Kampen theorem for a union along a connected bicollared subspace; the exact sequence $1 \to \pi_1(Z) \to \pi_1(W) \to G \to 1$ of a connected regular covering with deck group G; Poincaré–Lefschetz duality and universal coefficients, with the Euler characteristic as an alternating sum of Betti numbers over any field, multiplicative for bundles with compact fiber and additive along a common boundary; the index formula $\det(\mathrm{Gram}\, S) = [L : S]^2 \det(\mathrm{Gram}\, L)$ for a finite-index sublattice, with the unimodularity of the intersection form on H_2/Tors; Wu's formula $\langle w_2, x\rangle = x \cdot x$ and its consequence $\langle w_2(X), [F]\rangle = \chi(F) + F \cdot F \pmod 2$ for an embedded oriented surface; Novikov additivity of the signature; asphericity of bundles with aspherical fiber and base; the adjunction formula $K \cdot S + S \cdot S = 2g - 2$ for a symplectic surface; and the isotopy of any two orientation-preserving smooth embeddings of B^4 in a connected 4-manifold [24], which together with the amphichirality of the figure-eight knot makes smooth sliceness in a closed 4-manifold a diffeomorphism invariant.

The named theorems are the following.

- **Freedman** [9]: closed simply connected topological 4-manifolds are classified up to homeomorphism by the intersection form and the Kirby–Siebenmann invariant; every unimodular form is realized, uniquely if even (with $\mathrm{KS} \equiv \sigma/8$), and by exactly two manifolds if odd, one per value of KS.
- **Hambleton–Kreck** [10, 11, 12], in the form of [13, Theorem 5.1]: "Closed, oriented topological 4-manifolds with finite cyclic fundamental groups are classified up to homeomorphism by $\pi_1(M)$, q_M, the w_2-type, and $\mathrm{KS}(M)$," where q_M is the form on H_2/Tors and the w_2-type is (II) for spin M.
- **Thurston** [16] and the paper's Thurston-type form ([1], proofs of Proposition 1.3 and Lemma 8.2; [2], proof of Theorem 8): a closed surface bundle over a closed surface with homologically essential fiber is symplectic, and for $R \cup_\sigma R$ the form can be chosen positive on the fiber F and on the closed section $\widehat{\Gamma}$, with the surgery tori Lagrangian.

- **Luttinger surgery** [8, 3]: surgery on a Lagrangian torus yields a symplectic manifold, agreeing with the original outside the surgered neighborhood and with unchanged Euler characteristic.

- **Symplectic Kodaira dimension** [15], as summarized in [14, p. 1]: κ is defined from a minimal model and "is independent of the choice of symplectic form ω"; a minimal symplectic 4-manifold has $\kappa = -\infty$ iff it is rational or ruled, and rational and ruled 4-manifolds have $\pi_2 \neq 0$. **Ho–Li** [14, Theorem 1.1]: "The Luttinger surgery preserves the symplectic Kodaira dimension."

- **Symplectic Thom** [17]: an embedded symplectic surface in a closed symplectic 4-manifold is genus-minimizing in its homology class; and the construction ([1], proof of Lemma 4.3, after [20]) of a connected symplectic unbranched k-fold cover of a square-zero symplectic surface inside its tubular neighborhood, representing k times its class.

- **Klug** [21, Theorem 2]: "Let X^4 be a smooth compact connected oriented 4-manifold with ∂X an integer homology sphere. Let F^2 be an orientable characteristic surface with connected boundary that is properly embedded in X. Then $\mathrm{Arf}(F) + \mathrm{Arf}(\partial F) = (\sigma(X) - [F]^2)/8 + \mu(\partial X) \pmod 2$." For spin X, characteristic means vanishing in $H_2(X, \partial X; \mathbb{Z}/2)$. **Levine** [22]: $\mathrm{Arf}(K) = 0$ iff $\Delta_K(-1) \equiv \pm 1 \pmod 8$.

- **Slice disks and the 0-trace** ([2], proof of Theorem 1): a smooth slice disk with Seifert-framed normal bundle embeds the simply connected 0-trace, in which a capped Seifert surface is a closed square-zero surface generating H_2; the framing condition is automatic when the relative H_2 is torsion; the figure-eight knot has genus 1.

- **Lidman–Piccirillo's constructions** [2]: σ is a free orientation-reversing bundle involution (Lemma 4); the sections from the fixed points of φ_0 survive into V, their boundary circles are preserved by σ, and $R \cup_\sigma R$ is a closed genus-2 bundle over a closed genus-2 surface on which the surgeries act by Luttinger surgery on four disjoint tori missing F and $\widehat{\Gamma}$ (Lemma 6, proof of Theorem 8); $W = V/\sigma$ is a closed smooth oriented 4-manifold, double covered by Z, and is spin by the hyperbolic-pair argument of Lemma 7, whose hypotheses are the homology of V; the regluing diffeomorphism f of $S^3_0(Q)$ exists, and with $\mathcal{A} \subset S^3_0(Q)$ the framed annulus joining $\partial\Gamma$ to its image under $f \circ \sigma$, the closed surface $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma$ has odd square and meets F once (proof of Theorem 2); and the Floer-theoretic input, Lemmas 9 and 10 with the vanishing of all mixed invariants of $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ (and of its sum with any homology 4-sphere) along both square-zero lines, via splittings over $S^2 \times S^1$ (proof of Theorem 2, using [18, 19]).

- **Kawauchi's manifold** [23, 2]: B is a closed smooth oriented spin 4-manifold with $\pi_1(B) = H_1(B) = \mathbb{Z}/2$ and $b_2(B) = 0$, in which the figure-eight knot is smoothly slice.

The chain

Throughout, F is a fiber of R disjoint from the surgery tori, Γ a section through a fixed point of φ_0, $\widehat{\Gamma} = \Gamma \cup_\sigma \Gamma \subset Z$ and $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma \subset Z''$ its closures, $\mathcal{A}$ the framed annulus above. The finite calculations cited are those of the chain's certificate.

Proposition A.1. $H_1(V) = 0$, $H_3(V) = 0$, $H_2(V) = \mathbb{Z}\langle F \rangle$ with $F \cdot F = 0$, and V is spin. Moreover $\pi_1(Z) = \pi_1(Z'') = 1$.

Proof. Wuebben's Lemma 4.2 and the two filling relations give $H_1(V) = 0$ independently of simple connectivity; the certificate also implies it. The boundary $S^3_0(Q)$ is connected, so $H^1(V, \partial V) = 0$

and hence $H_3(V)=0$, and $H_1(V,\partial V)=0$ so $H_2(V)$ is torsion-free. $\chi(V)=\chi(R)=\chi(F)\chi(\text{base})=(-2)(-1)=2$, unchanged by the surgeries, and $b_0=1$, $b_1=b_3=b_4=0$ give $b_2(V)=1$. A section is a properly embedded surface meeting F once, so $[F]$ is primitive and generates. $F\cdot F=0$ since a fiber has trivial normal bundle. $H_2(V;\mathbb{Z}/2)=\mathbb{Z}/2\langle F\rangle$, w_2 is detected on it, and $\langle w_2,F\rangle=\chi(F)+F\cdot F=-2+0\equiv 0$. Finally Z and Z'' are unions of two copies of V along the connected bicollared boundary, so by van Kampen each fundamental group is a quotient of $\pi_1(V)*\pi_1(V)=1$. This is the one downstream step where simple connectivity of V, rather than only of Z, is needed: the regluing twists the amalgam. □

Proposition A.2. $H_2(Z)=\mathbb{Z}\langle F,\widehat{\Gamma}\rangle$ with intersection form the hyperbolic form H; Z is closed, simply connected, spin, with $b_2=2$ and signature 0; and Z is homeomorphic to $S^2\times S^2$.

Proof. $\chi(Z)=2\chi(V)-\chi(S^3_0(Q))=4$ and $\pi_1(Z)=1$ give $b_2(Z)=2$ with $H_2(Z)$ torsion-free. $F\cdot F=0$, $F\cdot\widehat{\Gamma}=1$ (a fiber meets a section once), and $\widehat{\Gamma}\cdot\widehat{\Gamma}=0$ by the certified self-intersection computation in the peripheral section of the main paper. The Gram determinant is -1 and the form is unimodular, so the index formula makes $(F,\widehat{\Gamma})$ a basis and the form is H. H is even, so $w_2(Z)=0$ by Wu's formula. For even forms $\mathrm{KS}\equiv\sigma/8=0$ is determined, so Freedman's classification gives $Z\cong S^2\times S^2$. Note that spinness of Z is obtained here from the certified $\widehat{\Gamma}\cdot\widehat{\Gamma}=0$; the paper's route through the spin quotient W is not needed for this step. □

Proposition A.3. Z is a closed symplectic 4-manifold in which F and $\widehat{\Gamma}$ are symplectic surfaces of genus 2 and square 0, and Z is not diffeomorphic to $S^2\times S^2$.

Proof. $R\cup_\sigma R$ is a closed surface bundle whose fiber is homologically essential ($F\cdot\widehat{\Gamma}=1$), so it carries the Thurston-type form, positive on F and $\widehat{\Gamma}$. The surgery tori are Lagrangian for it and the surgeries are the certified Luttinger surgeries (the audit-model theorem together with S1–S4 of the main paper), and Luttinger surgery preserves the form away from the surgered neighborhoods, which miss F and $\widehat{\Gamma}$. For the non-diffeomorphism: $R\cup_\sigma R$ is aspherical, hence minimal (an exceptional sphere would be null-homotopic, hence of square 0, not -1) and neither rational nor ruled (those have $\pi_2\neq 0$), so $\kappa(R\cup_\sigma R)\neq-\infty$; Luttinger surgery preserves κ, so $\kappa(Z)\neq-\infty$. If $\psi\colon Z\to S^2\times S^2$ were a diffeomorphism, composing with a reflection of one factor if necessary makes it orientation-preserving; ψ^* of a product form is then a symplectic form on Z inducing its orientation with $\kappa=\kappa(S^2\times S^2)=-\infty$, contradicting the independence of κ from the form. □

Propositions A.2 and A.3 are Theorem A.

Proposition A.4. W is a closed oriented smooth spin 4-manifold with $\pi_1(W)=\mathbb{Z}/2$, $b_2(W)=0$, signature 0, and $\mathrm{KS}(W)=0$; and W is homeomorphic to B.

Proof. $Z\to W$ is a connected 2-fold covering, so $|\pi_1(W)|=2$. $\chi(W)=\chi(Z)/2=2$ and $b_1(W)=0$ give $b_2(W)=0$, hence signature 0. More precisely, $H_1(W;\mathbb{Z})=\mathbb{Z}/2$ and Poincaré duality give $H_3(W;\mathbb{Z})=H^1(W;\mathbb{Z})=0$. The integral universal-coefficient sequence and duality then identify

$$H_2(W;\mathbb{Z})\cong H^2(W;\mathbb{Z})\cong \mathrm{Ext}(H_1(W;\mathbb{Z}),\mathbb{Z})=\mathbb{Z}/2,$$

because $H_2(W;\mathbb{Z})$ has rank zero. Thus the vanishing of b_2 hides one torsion class that is used in the sliceness argument below. The manifold W is spin by Lidman–Piccirillo's Lemma 7, whose hypotheses Proposition A.1 supplies, and $\mathrm{KS}(W)=0$ since W is smooth. W and B are then closed oriented topological 4-manifolds with the same Hambleton–Kreck invariants: $\pi_1=\mathbb{Z}/2$, the zero form on $H_2/\mathrm{Tors}=0$, w_2-type (II), $\mathrm{KS}=0$. □

Proposition A.5. No nonzero square-zero class in $H_2(Z;\mathbb{Z})$ is represented by a smoothly embedded torus.

Proof. For $k \geq 1$ the connected symplectic k-fold covers of F and of $\widehat{\Gamma}$ represent kF and $k\widehat{\Gamma}$ and have genus $k+1$, since $\chi = -2k$; by the symplectic Thom theorem these genera are minimal, and reversing orientation covers $k < 0$. Since $(aF + b\widehat{\Gamma})^2 = 2ab$, every nonzero square-zero class is kF or $k\widehat{\Gamma}$ with $k \neq 0$, of minimal genus $|k| + 1 \geq 2$. $\square$

Proposition A.6. The figure-eight knot is not smoothly slice in W.

Proof. Let D be a smooth slice disk in $W^\circ = W \setminus B^4$. The group $H_2(W^\circ, \partial; \mathbb{Z}) \cong H^2(W) \cong \mathbb{Z}/2$ is torsion, so the relative self-intersection of D is 0 and its framing is the Seifert framing. If $[D, \partial D] = 0$ in $H_2(W^\circ, \partial; \mathbb{Z}/2)$, then D is characteristic in the spin manifold W°, and Klug's theorem with $\mathrm{Arf}(D) = 0$, $\sigma(W^\circ) = 0$, $[D]^2 = 0$, $\mu(S^3) = 0$ forces $\mathrm{Arf}(4_1) = 0$; but $\Delta_{4_1}(t) = -t^2 + 3t - 1$ gives $\Delta_{4_1}(-1) = -5 \equiv 3 \pmod 8$, so $\mathrm{Arf}(4_1) = 1$.

It remains to make the other branch explicit. With $\mathbb{Z}/2$ coefficients, the long exact sequence of $(W^\circ, \partial W^\circ)$ and $H_2(S^3) = H_1(S^3) = 0$ gives

$$H_2(W^\circ; \mathbb{Z}/2) \xrightarrow{\cong} H_2(W^\circ, \partial W^\circ; \mathbb{Z}/2),$$

and Mayer–Vietoris for $W = W^\circ \cup_{S^3} B^4$ gives $H_2(W^\circ; \mathbb{Z}/2) \cong H_2(W; \mathbb{Z}/2)$. Consequently, if $[D, \partial D] \neq 0$, capping D by a Seifert surface in the removed ball produces a torus $T \subset W$ with $[T] \neq 0$ in $H_2(W; \mathbb{Z}/2)$. The Seifert framing gives $T \cdot T = 0$, equivalently the 0-trace $X_0(4_1)$ embeds in W.

The inclusion $i : X_0(4_1) \hookrightarrow W$ lifts through the connected cover $p : Z \to W$, because $\pi_1(X_0(4_1)) = 1$. For a chosen lift $\widetilde{i}$ one has $p \circ \widetilde{i} = i$; hence $\widetilde{i}$ remains an embedding and T lifts to a torus $\widetilde{T}$ on which p has degree one. Therefore

$$p_*[\widetilde{T}] = [T] \neq 0 \quad \text{in } H_2(W; \mathbb{Z}/2).$$

In particular $[\widetilde{T}] \neq 0$ in $H_2(Z; \mathbb{Z}/2) = H_2(Z; \mathbb{Z}) \otimes \mathbb{Z}/2$ (as $H_1(Z) = 0$), and a class with nonzero mod-2 reduction is nonzero in the torsion-free $H_2(Z; \mathbb{Z})$; and $\widetilde{T} \cdot \widetilde{T} = T \cdot T = 0$ because p is a local diffeomorphism. This contradicts Proposition A.5. $\square$

Proposition A.4, Proposition A.6, the sliceness of the figure-eight knot in B, and the diffeomorphism invariance of sliceness give Theorem B.

Proposition A.7. Z'' is closed, simply connected, smooth, with $b_2 = 2$, signature 0, and odd intersection form $\langle 1 \rangle \oplus \langle -1 \rangle$; hence Z'' is homeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

Proof. $\pi_1(Z'') = 1$ and $\chi(Z'') = 4$ give $b_2 = 2$ with H_2 free. Both Z and Z'' decompose into the same two oriented copies of V along their common boundary, so Novikov additivity gives $\sigma(Z'') = \sigma(Z) = 0$, independently of the regluing map. $F \cdot F = 0$, $F \cdot \Gamma'' = 1$, and $\Gamma'' \cdot \Gamma'' = 2\ell + 1$ is odd; the Gram determinant is -1, so (F, Γ'') is a basis, and $E = \Gamma'' - \ell F$, $D = F - E$ satisfy $E \cdot E = 1$, $D \cdot D = -1$, $E \cdot D = 0$ for every $\ell \in \mathbb{Z}$: the form is $\langle 1 \rangle \oplus \langle -1 \rangle$. Of Freedman's two manifolds with this form the one with $\mathrm{KS} = 0$ — Z'' is smooth — is $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. $\square$

Proposition A.8. Z'' is not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

Proof. Z is symplectic with $|K_Z \cdot F| = 2$ by adjunction (F symplectic of genus 2 and square 0) and $H^2(V) = \mathbb{Z}$, so Lidman–Piccirillo's Lemma 9 gives $\Psi_{V,t} \neq 0$ for both spin^c structures with $|\langle c_1(t), F\rangle| = 2$. Z'' is $V \cup V$ along $S^3_0(Q)$ with t restricting to the non-torsion $s_\pm$, $b_2^+(Z'') = 1$, and span$\{F\}$ a square-zero line, so Lemma 10 gives a nonvanishing mixed invariant $\Phi_{Z'',\mathrm{span}\{F\},t}$. A diffeomorphism to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ would carry span$\{F\}$ to one of its two square-zero lines — in $\langle 1\rangle \oplus \langle -1\rangle$, $a^2 - b^2 = 0$ forces $a = \pm b$ — along each of which the manifold splits over $S^2 \times S^1$ and every mixed invariant vanishes. □

Propositions A.1, A.7 and A.8 are Theorem C, and the Conditional Consequences Theorem of the main paper is proved.

Three points where the chain's route differs from the paper's are worth recording. $H_1(V) = 0$ is taken from the certificate rather than from the paper's Lemma 4.2 and the surgery relations, which are needed only for the "forcing" direction of its Proposition 1.3. The intersection form of Z is identified from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$, so that Z is spin because its form is even; W spin still enters, for the Hambleton–Kreck invariants and for Klug's theorem. And the sliceness dichotomy is taken over $\mathbb{Z}/2$ throughout, which is exactly what makes the null-homologous case Klug's characteristic case and the lifted torus nonzero integrally.

B Trust boundary and data availability

Checker provenance and audit status

The verification was developed by Anthropic Claude models (Fable 5 and Opus 5) and OpenAI Codex (GPT-5 family) under the author's direction, with commit-level provenance and cross-model corrections retained in the public ledger. Mathematical responsibility rests with the human author; model provenance is not evidence of correctness.

Both filled-group checkers were produced as separate Python and Ruby implementations by OpenAI Codex. They share neither implementation code nor runtime libraries beyond their standard libraries, but they do share the published certificate specification and input format. Thus "independent" means implementation-independent, not independent human authorship or statistically uncorrelated errors. The downstream pair was separately implemented in Python and Ruby by Claude Fable 5. No theorem-critical checker has yet received an independent human line-by-line audit. The small source files, negative controls, and exact replay commands are public so that this remaining software-review task is explicit.

Inputs to the finite theorem and audit-model theorem

The finite theorem of the main paper has no mathematical input beyond its Certificate Soundness Theorem; its implementation trust rests on the two small checkers. The audit-model identification with the explicitly defined V_* rests on standard inputs of PL topology: ribbon thickening and bistellar traces; classification of compact surfaces; local-flatness and normal-block criteria; and simplicial fundamental-group presentations with Tietze theory. The regular-neighborhood and local-flatness conventions are those of [5, Chapters 2–4] and [6, Chapters 3–5]; the presentation moves are the standard ones of [7]. The standard Lagrangian-neighborhood theorem supplies the Weinstein charts used by the framing theorem. This note applies rather than reproves those general results.

Source comparison with Wuebben

The main paper's S1–S4 are a separate evidence kind, not external theorems and not machine certificates. They assert marked-fiber fidelity, marked-bundle fidelity, peripheral/member fidelity, and smooth-surgery fidelity. Runs 51–56 and 68–69, the independent extractors, mutation controls, and the parsed author-script comparison give substantial evidence for those readings. They cannot prove what the target author's prose and figures are intended to denote. The exact residual and the evidence for each item are recorded in `verification/notes/source_identification_assumptions_2026-08-29.md`. Only the conditional comparison $V_* \cong V$ and the downstream conclusions depend on S1–S4; the finite theorem and $\pi_1(V_*) = 1$ do not.

Inputs to the Conditional Consequences Theorem of the main paper

The twenty-five external theorems of Appendix A, each stated there with its hypotheses. The deep ones are Freedman's classification, the Hambleton–Kreck classification, the invariance theorems for symplectic Kodaira dimension (Li, Ho–Li), the symplectic Thom theorem, Klug's relative Rochlin theorem, and the Floer-theoretic lemmas of Lidman–Piccirillo with the Ozsváth–Szabó nonvanishing theorem behind them; the rest are elementary or are constructions of [2] that this note takes as stated. None is reproved. The repository's dependency ledger records the whole verification as an acyclic graph of 33 named external theorems, 37 machine certificates, 12 geometric arguments, one source-assumption node, and 21 derived claims ending in the three theorems, with every evidence file bound by hash. These are not competing counts: the 25/3/15/16 figures in Appendix A describe only the compact downstream chain, while the larger figures describe the project-wide ledger that also contains the model, source comparison, peripheral, framing, and PL verification.

Data availability

This verification targets Wuebben's arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

`16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a`

The target-version ledger in `TARGET.md` also records the Lidman–Piccirillo v1 digest. All permanent certificates, frozen inputs, replay scripts, hash manifests, run transcripts, and provenance records are contained, with clean-checkout replay instructions, in the tagged release `v2.0.1` of the repository:

`https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v2.0.1`

The preceding v2.0.0 core verification snapshot is permanently archived at

`https://doi.org/10.5281/zenodo.22169754`

with concept DOI

`https://doi.org/10.5281/zenodo.22169753`.

The `verification/` tree of that release is identical to that of the repository commit

`a5ff4d4a7e90a2f34279aa9bd6777f018a6ce312` its filled-group proof manifest has SHA-256

`668e6dac41ac04340c25ca1eb06272512e8ad92353d94c866730dfefd5878d63`

and its downstream-chain certificate has SHA-256

`05a6f0726666a5926ec4824714e49bb8b7c65437b8227cb40b1f0065fd406349`

The repository itself is public at

`https://github.com/VentiMath/exotic-s2xs2-verification`

The versioned release link above, rather than search-engine indexing, is the artifact to check. The raw completion histories from which certificates are compiled are reproducible build products and are omitted. The paper's own code is at `https://github.com/bwuebben/exotic-s2xs2`; nothing in the verification derives from it except where explicitly cited.

C Supporting computational history

The proof of the finite theorem of the main paper uses only the sealed three-generator chain. An earlier export of the same marked complement had four generators and 95 relators. Its four $n = 0$ certificates contain respectively 1123, 1075, 804, and 1504 retained records and reach the same verdicts, but the interpreter hash seed used to number its raw simplices was not recorded; its input Tietze trace therefore cannot be regenerated. It is retained as corroborating history and no ledger node depends on it.

Four adjacent $n = 1$ presentations replace Ax by $Ar^{-1} = Ax(rx)^{-1}$. They pass the same full-inventory certificate pipeline and check Wuebben's one-step re-indexing, but they are not used for the fixed $n = 0$ member. Likewise, the framing-robustness scan tests 96 cross-and-diagonal meridian shifts plus four zero-shift controls: 22 have derivation certificates and 78 have retained GAP session verdicts. All 100 were reported trivial. Since these counterfactuals do not distinguish the correct framing, the scan is explicitly auxiliary.

The implementation has both positive and negative calibrations. On the standard T^4 Luttinger example it returns the complement $\mathbb{Z}^2 \times F_2$ and the textbook quotient $\mathbb{Z}^4$; on a Baldridge–Kirk example it separates the two sign conventions by trace fingerprints 1 and 3. A 96-relator common core completes to an infinite cyclic group, and a mapping-torus control distinguished $H_1 = \mathbb{Z}^2$ from the $H_1 = \mathbb{Z}^3$ identity case, exposing a construction error that was then fixed. Both filled-group checkers reject a wrong identity root, an altered input relator, an altered rewrite trace, a wrong generator count, an incomplete inventory, and a duplicated certificate batch. These tests show that the pipeline neither collapses every input nor accepts every file; the positive soundness claim remains the Certificate Soundness Theorem of the main paper.

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